PlayDapp Lab

Play To Earn Report

NFT Game Analysis



Blockchain games that allow regular players to earn simply by playing

I. The Definition of a “Blockchain Game” and Current Situation

1. Trend of Monetization Through Game Service

Traditionally, publishing and direct sales of games was the way of making money within the game industry. This business model has strong historic examples, such as the US giant Atari, which in the 1970s generated $20 billion in sales with its Pong series, the classic two-player virtual table tennis game.

In the traditional model most profits were made from ‘content business’, activities such as leveraging game IP or direct sales of the game itself. As well as licensing deals to other established game companies. In the traditional model profits went directly to the service publishers and content creators.

Then in the early 2010s, a newer earning model developed. Professional gamers emerged, participating in official leagues that were opened mostly as marketing avenues for established publishers. The development of the ‘pro gamer’ coincided with the appearance of professional influencers. These professional influencers used personality-based content delivered via streaming platforms, such as twitch and YouTube Gaming to reach wider audiences, often with the help of content that was heavily sponsored or facilitated by publishers. Together, the rise of ‘pro gamer’ and streamer culture showed that individuals could profit by playing games or by utilizing game content.

In 2021 we are faced with a new paradigm, the era of ‘Play to Earn’. The blockchain game industry is rewarding gamers for playing games, allowing them to monetize game content by simply playing them.

Video game console/PC/Mobile Game (1970s ~)	e-Sports and Game Broadcast (2010s ~)	Blockchain Games (2020s ~)
Developers, Creators	Professional Gamers, Game Broadcasting by Influencers	Individual Gamers
Content Create & Sales	Profit from Game Contents	Game Content Consumption and Monetization
Direct Sales and Distribution Content business by utilizing game IP	League Championship Prize Profit from advertisement	Reward from Gameplay In game Tournament Reward

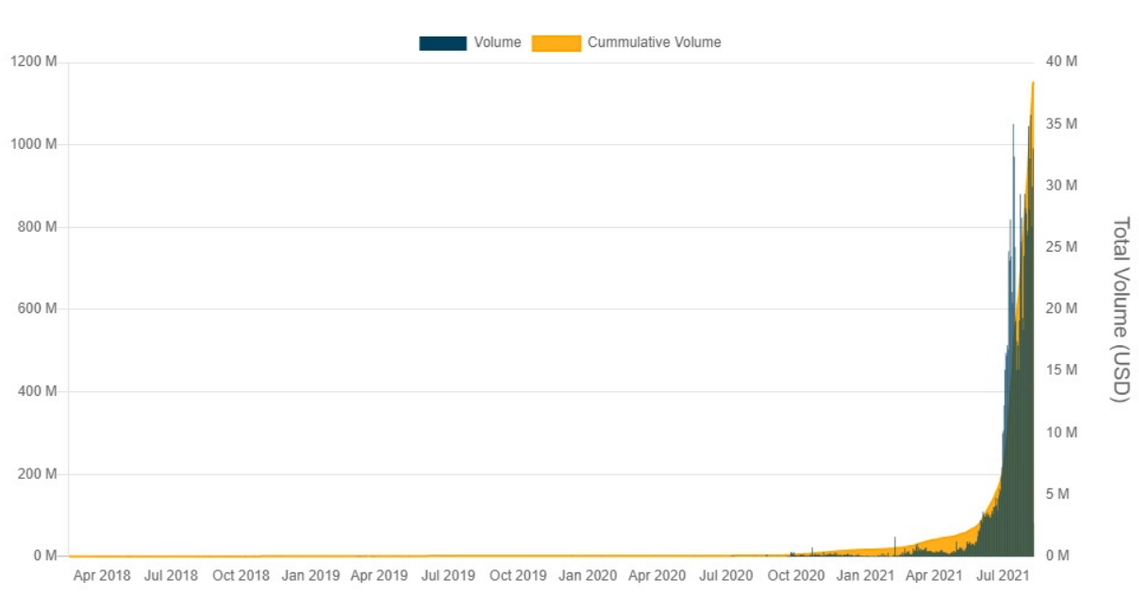
Game NFTs are leading the NFT market growth

Axie Infinity earned revenue of $400M in July 2021

2. Exponential Growth in the NFT-based Gaming Market

Digital art NFT, such as Beeple's 'Everydays: The First 5,000 Days' and the Crypto Punks and Bored Ape Yacht Club series led the growth of the NFT market from early last year to the first quarter of 2021. However, 2021's second quarter saw a sharp change, with Blockchain game-related projects driving NFT growth ever since. The second quarter of 2021 alone has seen the number of active wallets in blockchain games increase by 599% compared to the previous quarter. This dramatic growth is expected to continue into the third quarter of 2021.

SKY Mavis' Axie Infinity in particular, is gaining attention due to its historically unprecedented growth rate. As of August 9, 2021, Axie Infinity's DAU surpassed 800,000, and daily trading volume reached $33 million. In the third week of July alone, the company topped the Ethereum Foundation, which recorded $38 million in development team revenue and $32.8 million in revenue.



source : https://www.axieworld.com/

P2E expanded the pool of gamers beyond established Blockchain Popularity

II. Play to Earn Model the New Horizon for Blockchain Games

1. P2E (Play to Earn) an Innovative Business Model in the Game Industry

Axie Infinity's growth is closely linked with the Play to Earn (P2E) model. The P2E model attracts people from gaming's peripheries. The P2E business model empowers monetization by players by gameplay alone. We can see the success of this model in Axie, which sees more that 40% of it's users earning $1,000 per month under the P2E model.

This earning potential has hit new heights during the non-face-to-face era of Covid19. The new economic activity of playing and earning in games has created a 'new paradigm' moment.

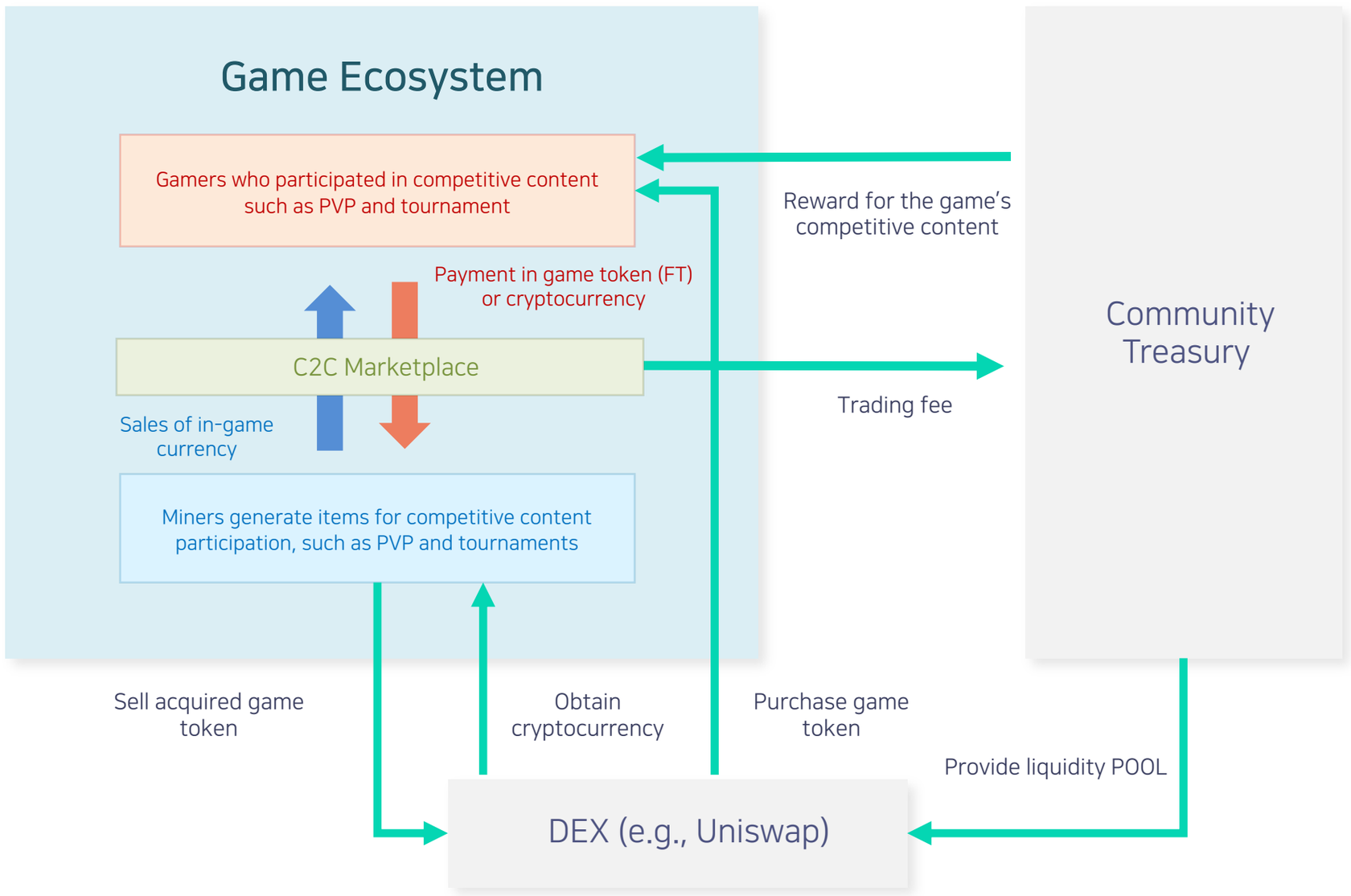
Pay to Play	Free to Play	Play to Earn
Traditional Video Game (1972~)	Freemium Game (2000~)	Play to Earn Game (2021~)
ATARI's Pong	Supercell's Clash of Clans	Blockchain Game
Pay to Play	Free to play	Play to Earn
Service Purchase (single product, monthly subscription)	Pay to win, In-game Customization, Ads	Presale, Miner, Tournament
Paying User, Core Gamer Specific	Non-Paying User, Target Audience Generalized Including Light Gamer	Target Audience Generalized Including Non-Gamer

Game revenue is returned to the gamers as a reward for their play

2. P2E (Play to Earn) Business Model Structure

At the core of the Play to Earn structure is the significant re-investment of the game's revenue into user activities. For Axie this reinvestment is about 95%. Alongside this reinvestment strategy the majority of key products required for gameplay are obtainable as in-game rewards or rewards for other participation, rather than the traditional model of selling game items.

Core game items earned by players are exchanged directly via a peer-to-peer (C2C) market or indirectly for game tokens. These game tokens can be exchanged for other cryptocurrencies on Decentralized Exchanges (DEX) or Centralized Exchanges (CEX). Under this new item acquisition method platform and game developers generate revenue via transaction fees incurred during each sale.



III. Analysis of a Play to Earn Game Ecosystem

Axie Infinity as a use case

Pay to Earn ecosystem
Use case :
AXS, SLP, Axie ways

1. Axie Infinity's Core Currency

Axie Infinity has three main currencies: Axie Infinity Shard (AXS) a governance token that enables the holder to become a decision maker in and owner of the Axie ecosystem. Holders of AXS also are entitled to a revenue share of transaction fees and AXS fees used for breeding.

Smooth Love Motion (SLP) is an essential currency used to produce Axies through breeding. SLP is not sold directly to players. It is instead only obtainable as rewards for PVE and PVP action.

Axies (NFTs) are used as an entry requirement as well as game character. Details on obtaining Axie NFTs and their unique usage are discussed in detail on part III-2.

	AXS (Axie Infinity Shard)	SLP(Smooth Love Potion)	Axie (NFT)
Token Types	ERC -20 / Ronin Token	ERC -20 / Ronin Token	ERC-721 / Ronin NFT
Usage	Axie Governance Token (Owner)	In-game play rewards and breeding	Game Character (Ticket)
How to Use	• Key decision making by staking and revenue distribution (TBA) • Use 2 AXS for each breeding • Use as a game/market payment	• Use for breeding (7 breeding per Axie, cost increases by 5 for each try)	• Requires at least 3 for exploration and battle • ETH version will not be used in the game from May 29th
How to Acquire	• Token sales and purchases from more than 80 exchanges Top tier content play rewards (PVP, etc.)	• PVE Exploration and PVP Rewards	• Presale and breeding Purchase from NFT Market
Trade (Sell) Methods	Sold on more than 80 exchanges	Binance or Uniswap	Sell internal & external NFT market
Price	$41.88	$ 0.243	$250~$2,450
Market Cap	$2,548,887,200	$131,774,424	N/A
Current Distribution Volume	60,907,500	541,173,609	9,106,199
Max. Issuance	270,000,000	No limitation	No limitation

* Key indicators as of July 30, 2021 (Land will be used to strengthen Axie, but it has not yet been deployed as a service)

2. Axie NFT

Axie NFT (hereinafter Axie) are the most important content in the game. With Axie no longer on general sale, (an initial batch of 3,888 Origin Axie were sold in April 2018), the only acquisition method is via breeding or player trade. Axie have three main uses:

Three meaningful uses for Axie NFT

- Ticket to prevent inappropriate account generating
- Used as key currency for game's core content
- Obtain additional energy to play

A. Entrant item

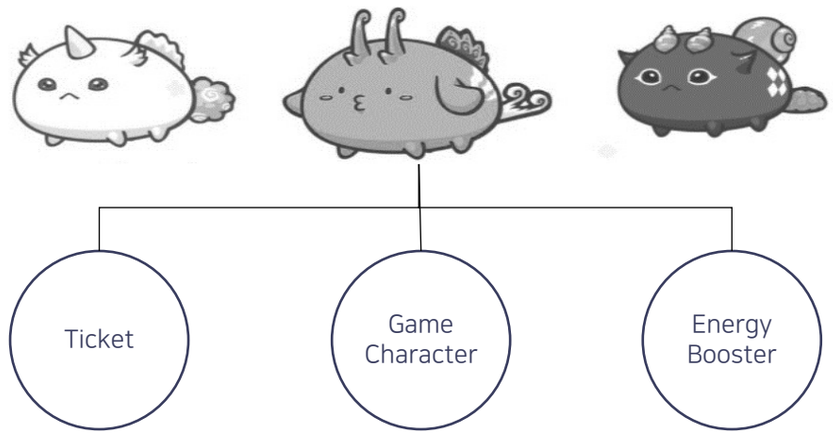
At least three Axies are required to play. The lowest price of an Axie currently trading at approximately $250. To begin participating in the game, an initial cost of approximately $750 is required, to cover three Axie purchases. This initial entry hurdle is designed to prevent game abuse via indefinite account generation to receive rewards.

B. Game Character

Axies are divided into nine classes, with three special classes. Each class is subdivided into six body parts: eyes, mouth, ears, horns, back, and tail. Each body type consists of three genes, defined as dominant (D), recessive (R1), and pyrogenic (R2). The combination of these genes determine which 'cards' (capabilities) are available when battling Axie. Due to the complexities in class characteristics numerous combinations of Axie are possible. This creates the need for highly strategic purchasing and breeding of Axie according to the players desired gameplay. Breeding focus, PVP or PVE focus all play a key role in determining a player's purchasing behavior.

C. Energy Booster

Player participation in PVE & PVP is reliant on energy. Overall energy and its recharging rate is determined by the number of owned Axies by the player. Those who are seeking to rank high in PVP will on average need 20 Axies to create maximum energy.



No. Axie	Max. Energy	Replenish Rate
3~9	20	Restore 5 energy every 6 hours
10~19	40	Restore 5 energy every 3 hours
20+	60	Restore 5 energy every 2 hours

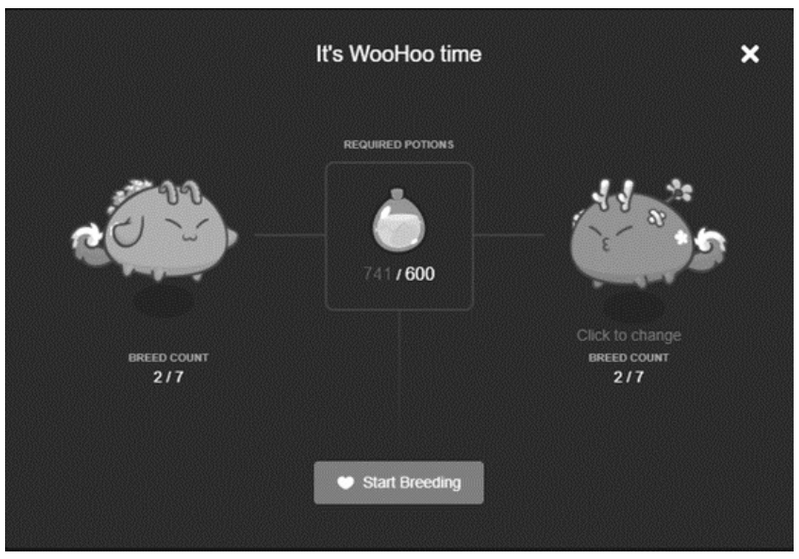
*Three main meanings of Axie NFT (left) and energy charge rate according to Axie count (right)

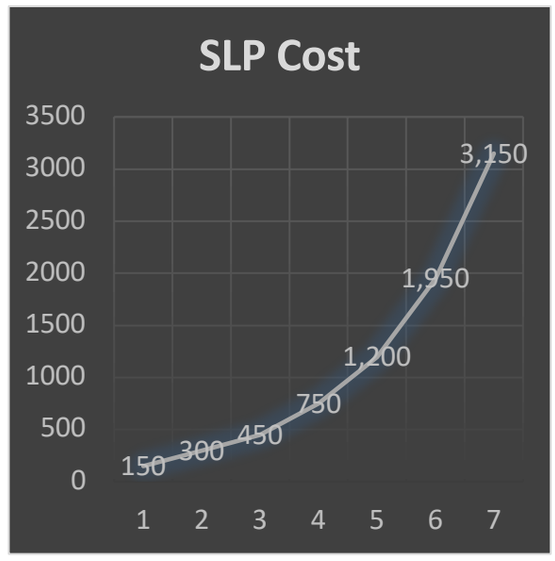
3. Breeding Systems and Core Currency SLP (essential to game's core content)

SLP is a core currency for breeding

As mentioned in part III-2, Axies are not sold separately. Currently, Axies are only created through breeding, which consumes AXS and SLP.

Two systems exist to prevent indiscriminate reproduction. The first is a maximum of seven breeding times per Axie. The second is an increase in the cost of SLP coins consumed when breeding. SLP cost increases with each successive breeding.





4. SLP's Virtuous Cycle in Game Economy

A virtuous cycle structure designed to use SLP to obtain Axie

It is important to draw attention to Axie Infinity's virtuous cycle between its currency SLP and the mechanic of breeding. In May 2021 Axie Infinity paid the players 150 SLP per day, then worth $15. This payment was a reward just for playing the game. As a result of this daily payment of SLP the player pool increased to receive the rewards. As the pool expanded, so did the number of people wanting and needing more Axies, due to the need for a minimum of three Axies to play.

Currently, the only way to increase your number of Axies is through breeding. SLP is a requirement of the breeding process. As SLP is used and the available pool decreases the value a player can charge for it rises. Currently the game pays the same 150 SLP daily log in reward as before, but over time the value of SLP has increased. In July 2021 the value of the daily 150 SLP reached $45 (three times its value in May 2021). This value increase will lead to the increase of the player pool once again, as they seek to farm the valuable SLP. The relation of SLP to breeding creates a virtuous cycle that will ultimately drive up the price of the SLP.

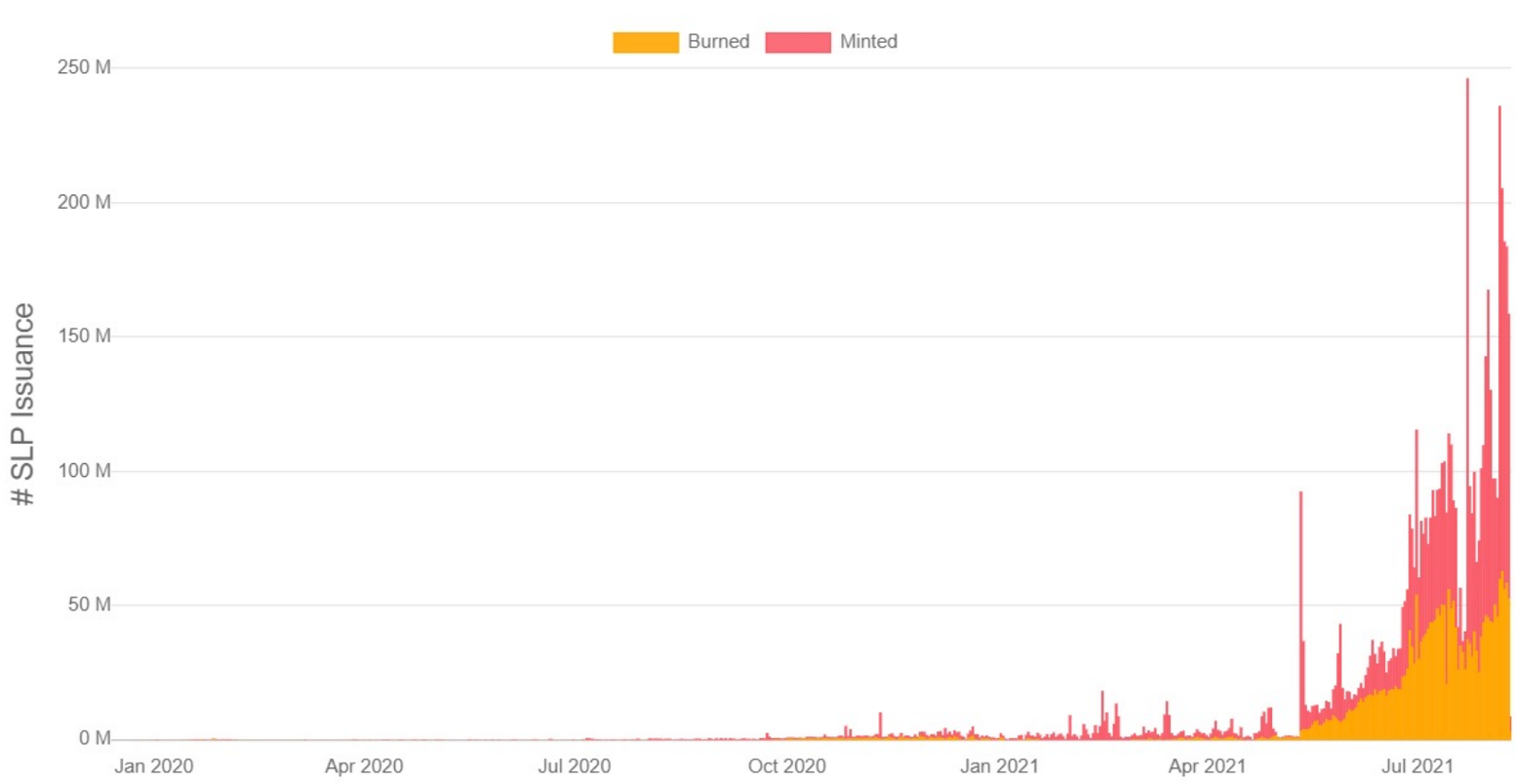


SLP's Circulation Adjustment – an important factor > Possibility of intentional platform intervention

5. The Effect of SLP Distribution on Game Ecosystem

An ever-increasing SLP price is always an advantage for the game ecosystem.

If SLP price is too high, the cost of generating Axies with breeding becomes inflated, as does secondary sales as the cost to purchase reflects the difficulty to breed similar Axies. This shared rising cost crates increased entry barriers and endangers game balance. Although between May and June the value of burned SLP was significantly lower than that of minted SLP the price of SLP has been stabilized as SLP minting has expanded since July. However, it is difficult to distinguish whether the increase in distribution volume was due to the large volume of new players who literally just tried the breeding without a thoughtful strategy or whether there was intentional intervention by the platform.



SLP price is closely related to ETH price changes

6. Price Correlation Between SLP and ETH

The graph below shows that SLP price movement is closely related to ETH price movement in addition to distribution volume.

On Axie's own Ronin platform, not only will the initial price notation be in wETH, but the Axie transaction will also be based on ETH. This is proof that the game ecosystem of Axie is affected by the blockchain market (especially the ETH price). This means that all ecosystem participants receive more than expected rewards, when including the ETH price increase factor. It also indicates that the service could be hit hard with significant Ethereum price fluctuations.

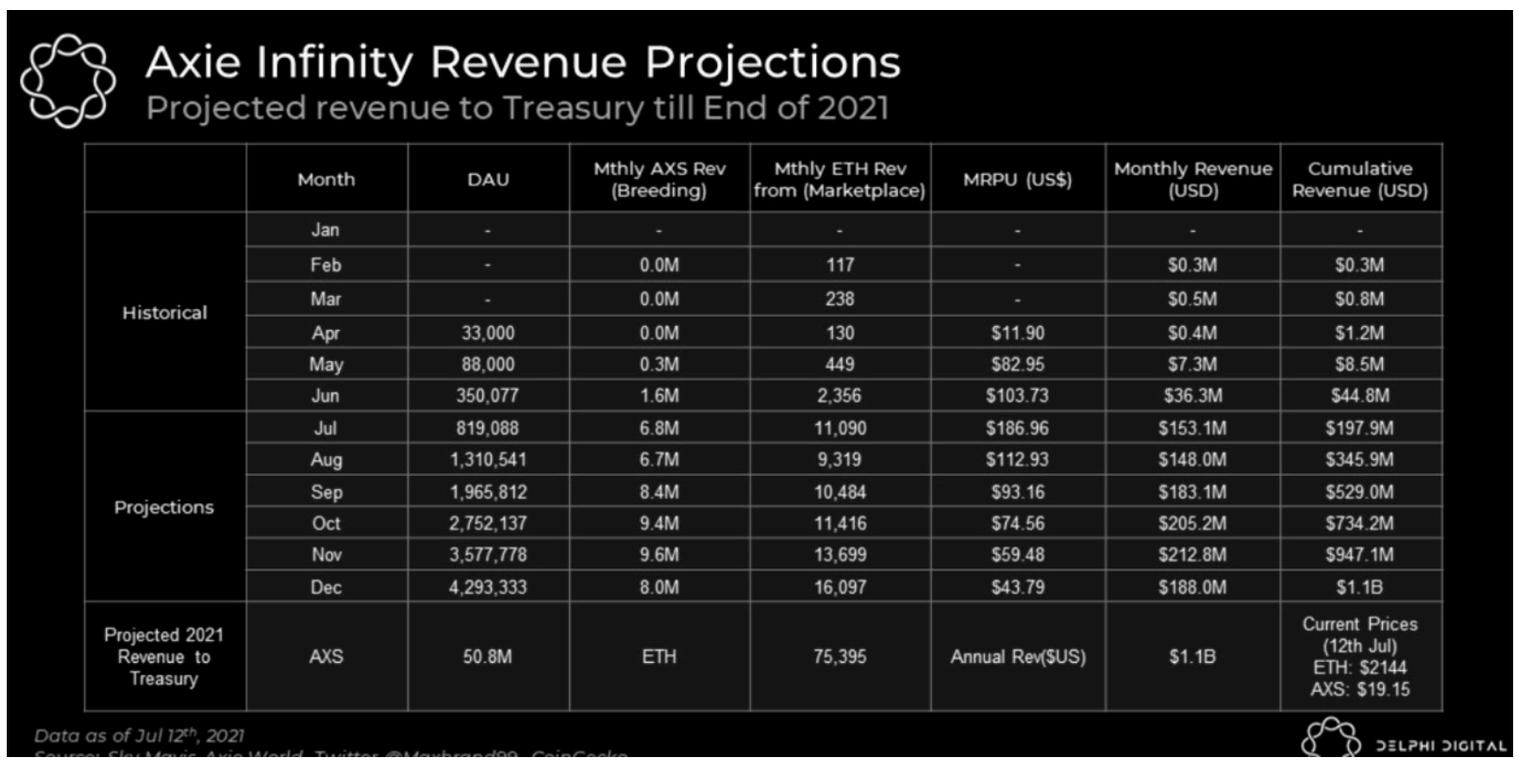


$2.8 million daily revenue Axie being accumulated

7. Current Sales Status and Forecasts

Axie Infinity's daily sales and MRPU are higher than other traditional games, with $2.8 million sales per day and a MRPU of $186. This is despite the fact that much of its revenue is returned to users. By July 2021, not only did cumulative sales exceed more than $200 million, but also sales volume in general and DAU are grew rapidly. However, as of the end of December 2021, the MRPU is targeted at $43, which is ¼ of current volume. This presumably indicates that Axie's team is now being careful about excessive consumption and compensation of Axie's major currencies. As well as distribution to players due to rising ETH and SLP prices. It's reasonable to expect action to be taken to resolve this issue.

** MRPU (Monthly Revenue Per User)*



	Month	DAU	Mthly AXS Rev (Breeding)	Mthly ETH Rev from (Marketplace)	MRPU (US$)	Monthly Revenue (USD)	Cumulative Revenue (USD)
Historical	Jan	-	-	-	-	-	-
	Feb	-	0.0M	117	-	$0.3M	$0.3M
	Mar	-	0.0M	238	-	$0.5M	$0.8M
	Apr	33,000	0.0M	130	$11.90	$0.4M	$1.2M
	May	88,000	0.3M	449	$82.95	$7.3M	$8.5M
	Jun	350,077	1.6M	2,356	$103.73	$36.3M	$44.8M
Projections	Jul	819,088	6.8M	11,090	$186.96	$153.1M	$197.9M
	Aug	1,310,541	6.7M	9,319	$112.93	$148.0M	$345.9M
	Sep	1,965,812	8.4M	10,484	$93.16	$183.1M	$529.0M
	Oct	2,752,137	9.4M	11,416	$74.56	$205.2M	$734.2M
	Nov	3,577,778	9.6M	13,699	$59.48	$212.8M	$947.1M
	Dec	4,293,333	8.0M	16,097	$43.79	$188.0M	$1.1B
Projected 2021 Revenue to Treasury	AXS	50.8M	ETH	75,395	Annual Rev($US)	$1.1B	Current Prices (12th Jul) ETH: $2144 AXS: $19.15

Source : https://www.coindesk.com/axie-infinity-token-price-doubles-in-two-days

PVE Miners aim for guaranteed profit at minimum cost

IV. Play to Earn Game Player Analysis

1. Miner's Gameplay and Rewards

A. Miner's PVE Gameplay and Rewards

Players as miners usually focus on buying the cheapest three Axies and obtaining 100% guaranteed Adventure PVE 100 SLP rewards. When initial Axie purchase costs are a burden, there is also a way to borrow Axie through the scholarship program and share a certain portion of SLP revenue (ranged at 25%~40% share of rewards).

- Quest: 50 SLPs, daily quests, attendance rewards, 10 Adventures wins, 5 Arena wins
- Adventure: up to 100 SLPs, PVE, and stage reward of SLP for each Map



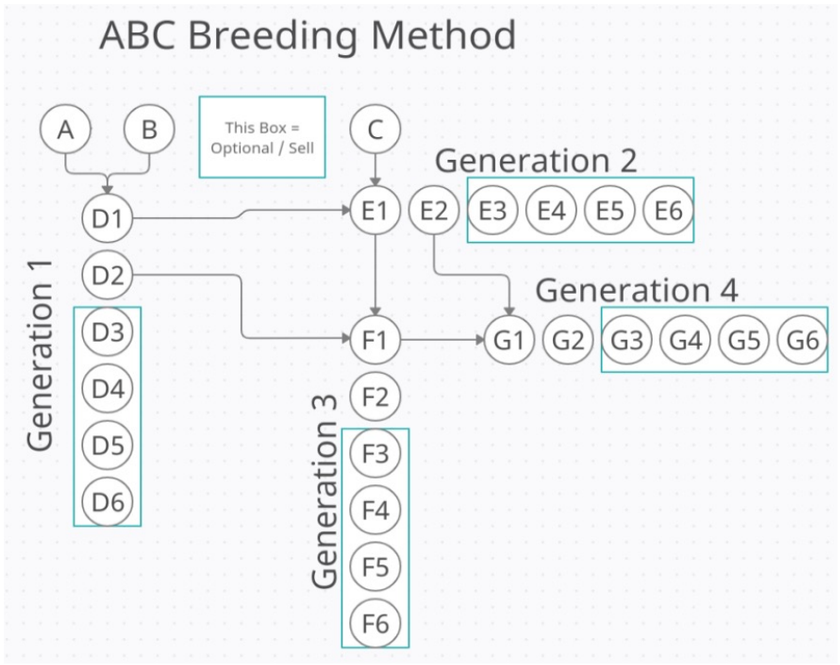
Map Stage	Reward for Win (SLP)
1~4	1
5~9	2
10~14	4
15~16	6
17~20	6~10
21~26	10~20

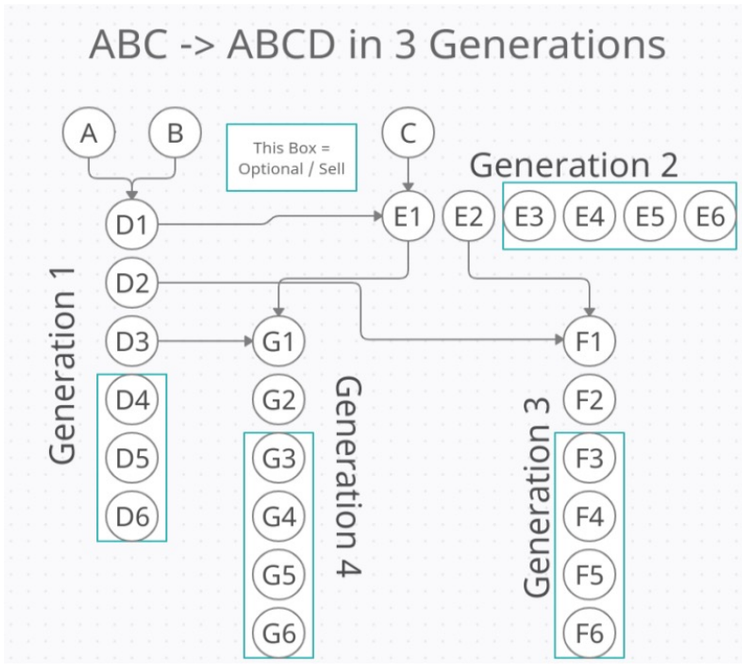
breeding Miners aim for constant revenue from Axie's breeding structure

B. Miner's Breeding and Rewards

After breeding, it takes 5 days for Axies to hatch from their egg form. Considering the current Axie and SLP costs, the most ideal structure to maximize profit is to sell Axies that are created from Axies that have been used for breeding less than twice. Due to the SLP consumption increase cost profitability is greatly reduced after more than 2 breedings.

Given that breeding between close family members (Parent-offspring breeding) is strictly prohibited, if managed correctly the first three Axies to hatch enable you to reproduce Axies into an infinite loop. Earning revenue as shown in the below diagram. However, you must remember that there is price volatility of Axie, SLP and AXS due to the nature of liquidity. The result is that you cannot assume the constant efficiency





Source : https://lynnyl.io/axie-infinity-breeding-guide/

2. Gameplay and Rewards

PVP Gamers aim for greater reward by utilizing the best Axie

A. PVP Gameplay and Rewards

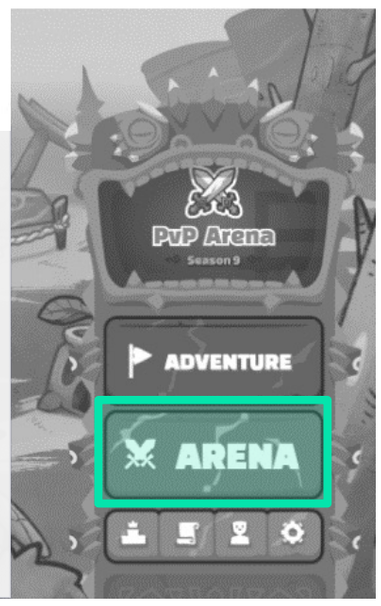
Users who engage primarily with gameplay aim to achieve high ranks in PVP. This results in huge AXS rewards. To do this, Players must spend their resources wisely. PVP rewards are compelling, the 1st prize for the 18th PVP season (ending SeP2Ember 19, 2021), is 222,500 AXS worth $13 million as of August 10th, which is significantly more than any E-Sports prize.



PVP Player : Gamer

AXS, SLP Rewards by achieving top rankings

- Season's 1st ranking prize was 225,000 AXS, and up to 2,000 SLP per day



- Arena : PvP combat, Reward varies depending on MMR (Match Making Rating)

MMR	Prize for the Win(SLP)
800 under	N/A
800~999	1
1,000~1,099	3
1,100~1,299	7
1,300~1,499	8
1,500~1,799	9
1,800~1,999	10
2,000~2,199	11
2,200 이상	12

In case of draw match,
the reward is halved for each respective party

Players continuous attempt to crossbreed to get the best Axies

B. Breeding Rewards

Having a powerful Axie is essential to achieve a high PVP rank. In order to obtain an Axie of pure bloodline with good characteristics, players must repeat the breeding process until finally getting a sufficiently powerful Axie. Breeding for secret classes is especially difficult. This is because a specific combination of two pure bloodline Axies is required. Breeding players are incentivized to continually seek ways to improve their limited breeding ruins, considering the drop off in potential profit after successive breeding.

Even successful breeding of pure-blooded Axies only has a 33% chance of creating a secret class.

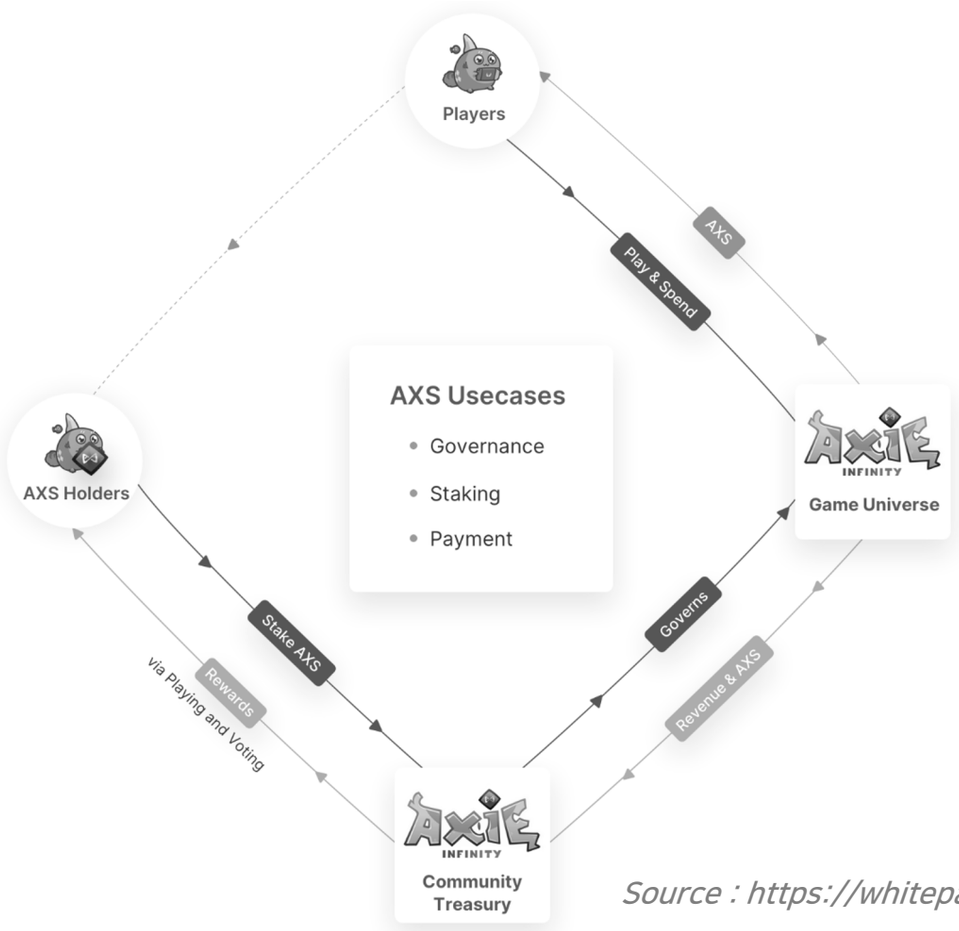
Dusk Class: 33% chance when breeding a pure Reptile with a pure Aquatic.
Mech Class: 33% chance when breeding a pure Bug with a pure Beast.
Dawn Class: 33% chance when breeding a pure Plant with a pure Bird.

Source : https://medium.com/axie-infinity/secret-classes-in-axie-infinity-129d0e69f1f9

3. Implications and Significance

The balance of “Time Cost” vs “Spend Cost” is deeply embedded in the game. “Time Cost” vs “Spend Cost” is already an industry standard. According to a 2011 report by Dr. Vili Lehdonvirta, about 400,000 Chinese worked for WOW’s (World of Warcraft’s) gold farming and power leveling workshop through brokers, and they were being paid approximately $150 a month in 2009. Axie’s balance allows for a healthy ecosystem between those who pay and those who put in the time.

Although the game industry is steadily growing in the non-face-to-face era, the financial benefits derived from this growth are enjoyed by game companies and distributors. With users, who contributed to the growth and development of the service not receiving the right compensation. Axie Infinity has proven the success of the new business model in that it shares game revenue (SLP) and platform growth rewards (AXS) to users, rather than monopolizing the rewards that occur in the game.



Source : https://whitepaper.axieinfinity.com/axs

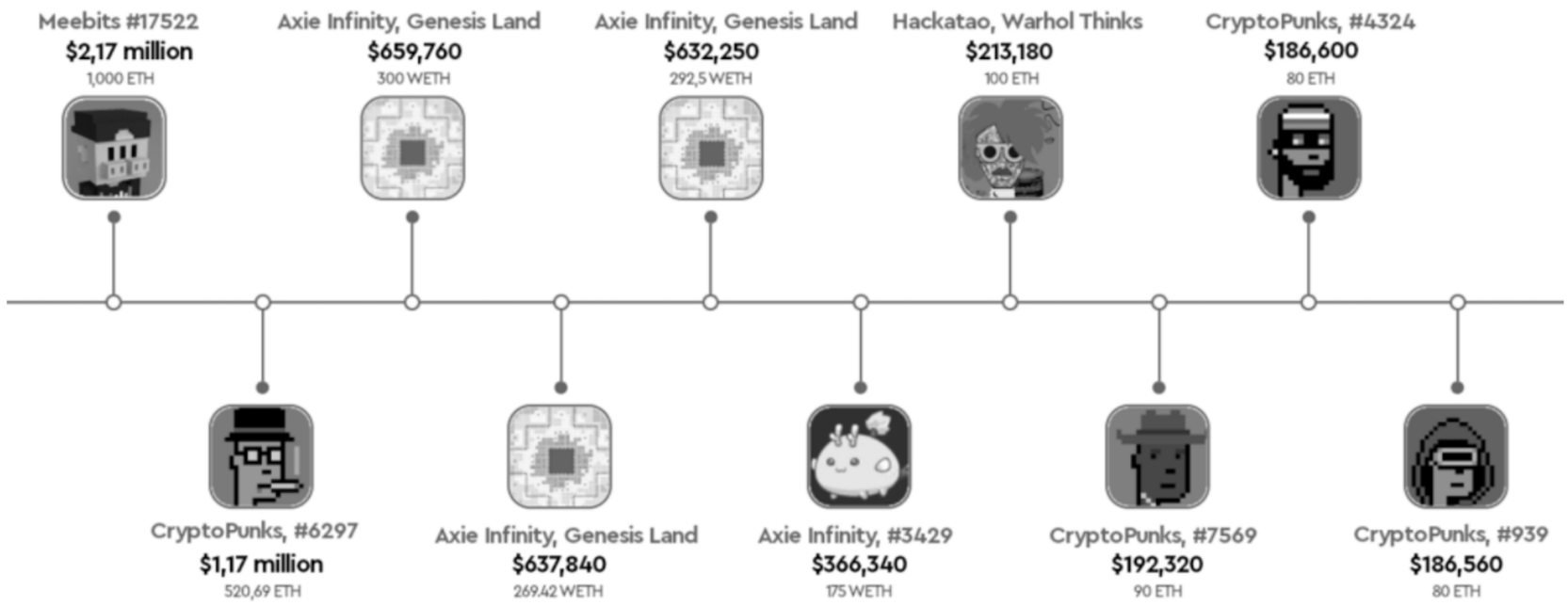
Questions resulting from their inhouse sidechain ‘Ronin’

V. Axie Infinity’s Limitations

1. Inhouse Sidechain, ‘Ronin’

Axie Infinity handles many transactions on the Ethereum sidechain ‘Ronin’. Created by SKY Mavis to improve its gaming services. However, it is difficult to ensure transparency and fairness, one of the most important elements of blockchain, given that the sidechain service is directly operated by the development entity (Which means it is an in-house product). For example, in February 2021, when Axie’s trading volume was low, Axie’s land sale achieved the highest NFT price ever, with $1.5 million being spent. However, as the deal was on SKY Mavis’ own Ronin (side-chain) platform, questions and doubts can be raised, over credibility.

Top 10 NFT Sales, on the week from the 5th to the 11th of July,
4 of the 6 top sales were Axie Infinity NFTs



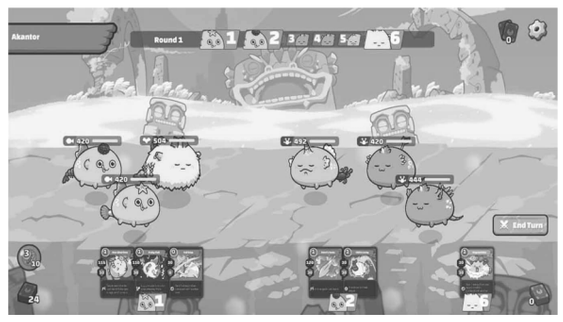
2. Axie’s Attractiveness as a Game?

Axie’s limitation as a Game

Axie may be loved by many, for its earning capabilities. However, if you look at it in purely traditional game service terms there are many areas of improvement. SKY Mavis’ 40 developers run a marketplace, service platforms as well as being the same developers tasked with creating games from scratch. Therefore, creating the sustained level of traditional game-like quality might become an overwhelming expectation for them. Due to this, inevitable long-term consequences may follow. When comparing the blockchain game ecosystem to traditional games, the amount of game content in PVP and PVE games is significantly unsatisfying.





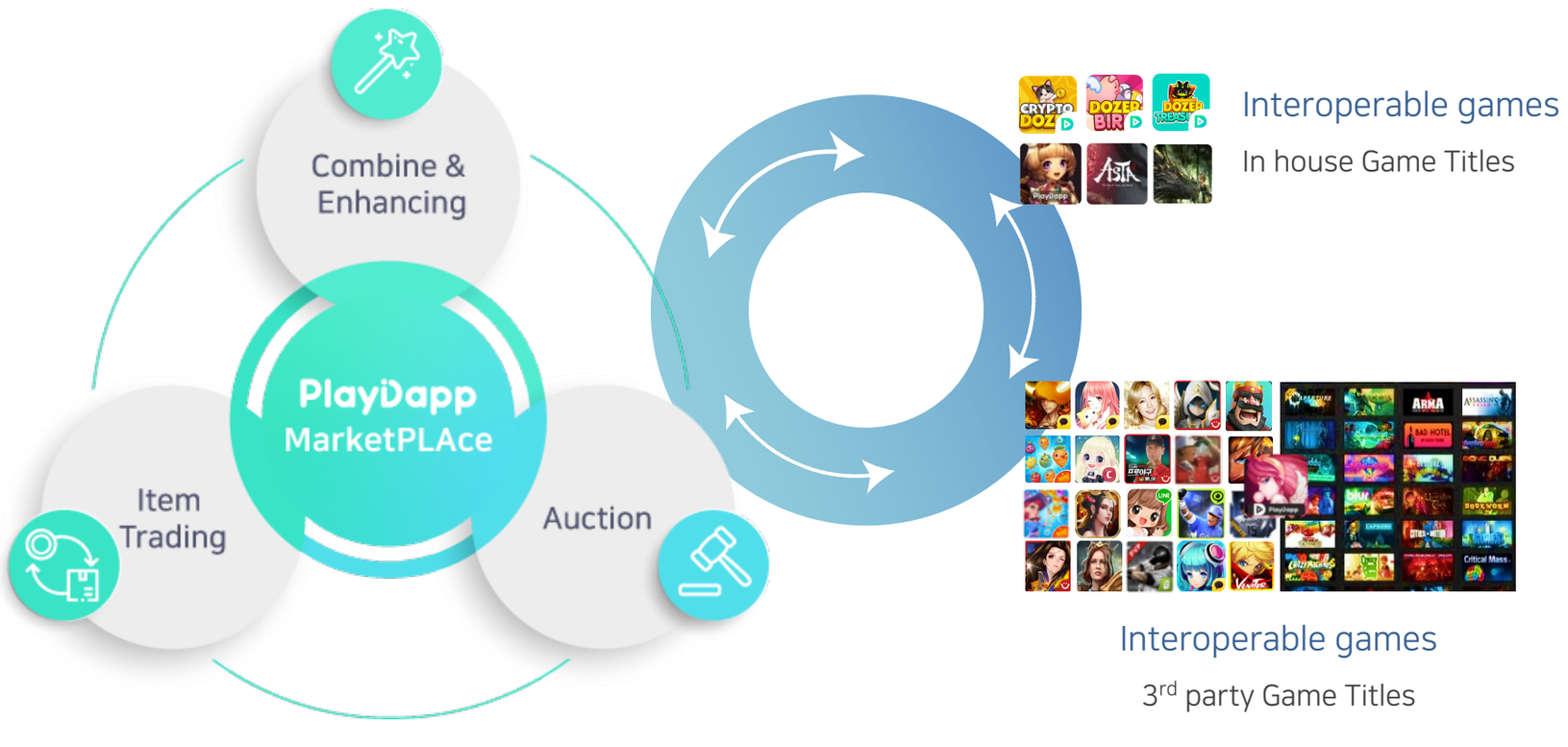


VI. PlayDapp Business Strategy

1. PlayDapp Marketplace

Onboard and service only proven games which increase the value of your digital assets

PlayDapp's vision is to create a world where digital assets are more useful and valuable for everyone. Users can freely trade items, characters and accessories on a blockchain-based C2C Marketplace. On PlayDapp’s marketplace users can take 100% ownership of their digital assets. As well as enhance, strengthen, and synthesize them into something new. Additionally, game items are not limited to use in a single game or intellectual property nor bound to one singular area. Instead, all can be operated interchangeably within the PlayDapp service ecosystem.

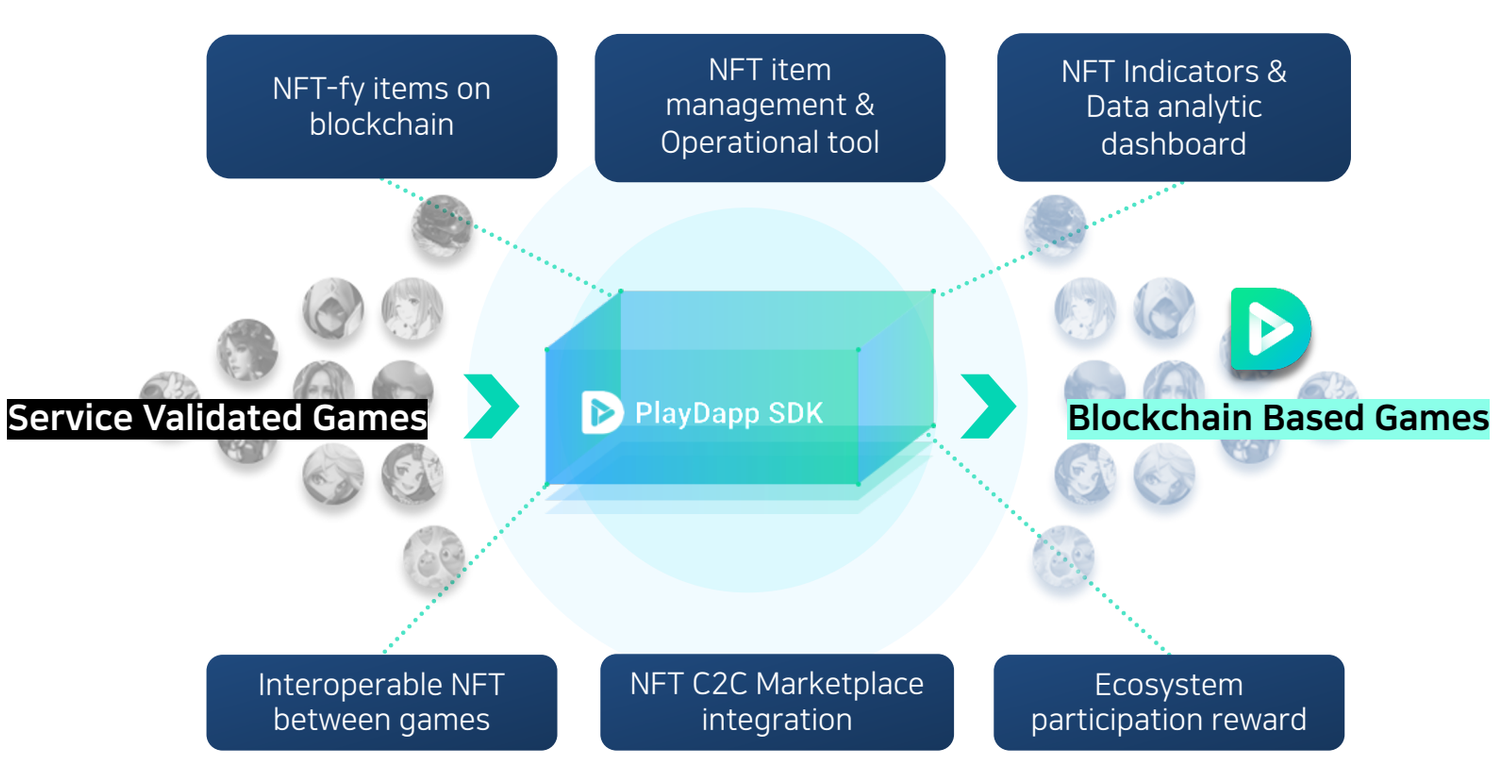


2. PlayDapp SDK

Transform the 3rd Party Game Items into Blockchain Based NFT

PlayDapp develops and provides a PlayDapp SDK to help game developers easily participate in the blockchain-based PlayDapp service ecosystem. The SDK helps game developers or companies who have difficulty building blockchain infrastructure to connect their assets or desired digital items with a simple system integration. Once itemized as a blockchain-based NFT, it can be interoperated between other various games on the PlayDapp ecosystem. This can create synergy between games and facilitate user acquisition through cross-promotion. In particular, the value of NFT can be increased by increasing the usage of the game items.

PlayDapp is currently servicing a Game specific NFT marketplace



Transaction transparency
Decentralized service operation

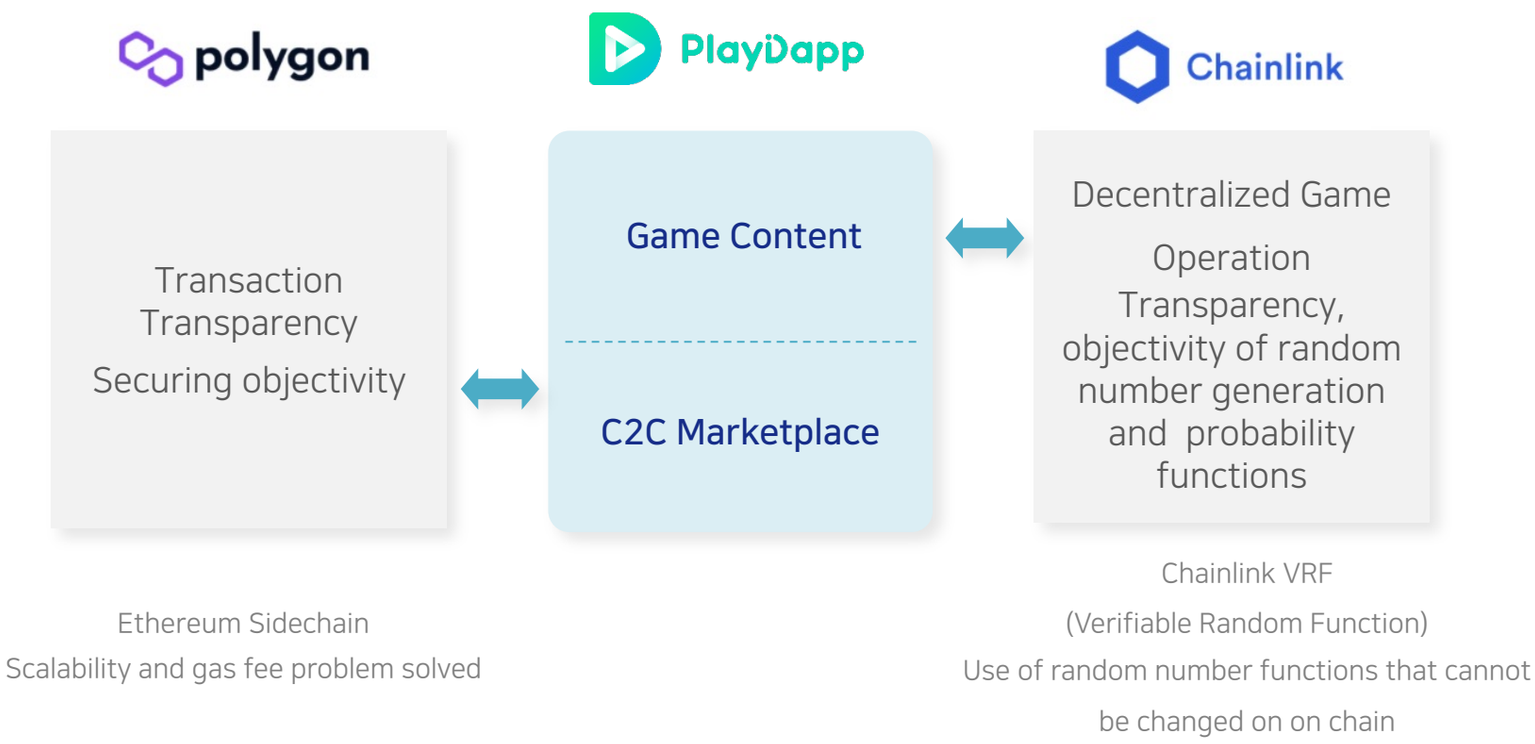
Transparency in random number probability and objectivity

3. Using a proven global top-tier blockchain technology platform

Providing a decentralized service operation system that is transparent, and objective

PlayDapp works with established global top-tier blockchain technology-based foundations. By utilizing their blockchain-based services PlayDapp can create trusted and reliable operation systems. PlayDapp aims to decentralize service operations and release key indicators transparently.

- To address Ethereum's scalability issues, we work with the 3rd party provider, polygon networks (a Layer 2 provider) rather than with our own sidechain to serve on EVMs.
- To maintain fairness and open probabilities for random values when merging items and upgrading NFT items PlayDapp uses VRF (Verifiable Random Function), A 3rd party product serviced by Chainlink. Removing the burden of proving fairness by PlayDapp's inhouse services.



PlayDapp develops under strict project operation protocols, Proof of Product and Proof of Service as a priority. Through this, it is listed on Coinbase, Upbeat, etc.

coinbase
https://www.coinbase.com/price/playdapp

UPbit
https://www.upbit.com/exchange?code=CRIX.UPBIT.KRW-PLA

4. PlayDapp Game Portfolio

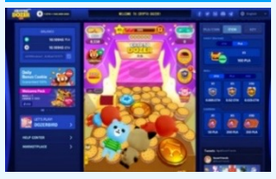
Game titles across various genres, including casual, RPG, Metaverse, and tournament

Multiple game titles that can be Play to Earn

PlayDapp services various genres of games developed in-house and by 3rd party developers. Most games are released on Google Play Store and Apple App Store. Game items can be made into blockchain-based NFT items through the PlayDapp SDK and Naver GamePot API.

Play-to-earn (P2E) models can be applied to proven RPG games via game guilds and communities, game content like boss raids, infinite tower raids, weapons, powerful runes and heroes. This combination creates highly enjoyable blockchain based game services, rather than simple, monotonous gameplay.

CryptoDozer



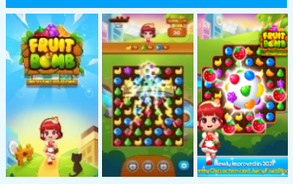
DozerBird



Sweet Cookie



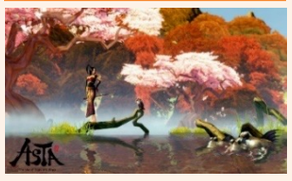
Fruit Bomb



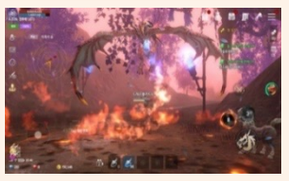
CASUAL

RPG

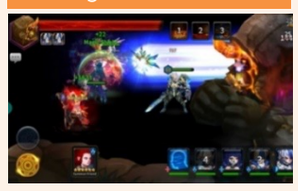
ASTA (MMO RPG)



Dragon Blood (MMO RPG)



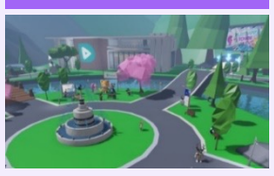
Along with the Gods



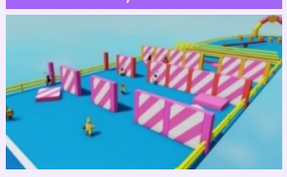
Blade Waltz



PLAYDAPP TOWN

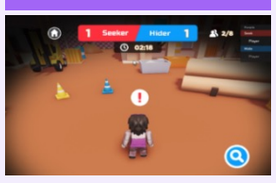


Playz Run

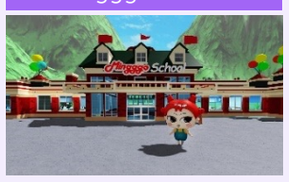


Metaverse

Hide & Seek

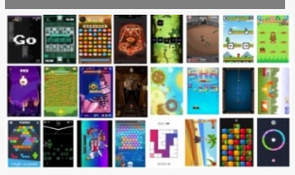


Mingggo School



Tournament

30+ Instant Games



Town Rangers : TPS



Interoperated NFT items that can be used between games

5. Interoperable NFTs

Increase the Value of Game Services with Interoperable NFT by Using PlayDapp IP

If blockchain technology is introduced into existing games, NFT items can be operated interoperably and act as a factor that expands game services. No matter how interesting a game is, there is a Product Life Cycle (PLC). When a user wants to play a new game, Interoperable NFT, which utilizes PlayDapp's own IP, can serve as a reward for previous gameplay and a natural cross-promotion of new games. This is a means of minimizing marketing costs incurred when recruiting users, and it can further strengthen the PlayDapp ecosystem by providing more diverse benefits to users.



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